Network Data Analysis

Coursework Report

Module: 7CUSMNDA Network Data Analysis

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Code Repository:

https://github.com/Shui-Zhou/kcl-network-data-analysis-final

Part 1: Wikidata Editor Co-Comment Networks

Wikidata, the structured knowledge base maintained by the Wikimedia Foundation, relies on community-driven editorial processes including property proposals, deletion reviews, and bot task requests. Each process generates discussion threads where editors participate through comments. In this part, I construct and analyse co-comment networks from three Wikidata discussion datasets, representing different scales and community dynamics. I characterise these networks using small-world and scale-free metrics, and simulate trolling propagation to assess community vulnerability to disruptive behaviour.

Task A: Network Construction

Dataset Selection and Edge Definition

Three Wikidata discussion datasets were selected to represent different scales and community types, each containing three columns: thread_subject, username, and page_name.

REQUEST_FOR_DELETION (large, 648,637 records): Discussions about proposed deletions of Wikidata items. This is the largest dataset by far, involving 9,935 unique editors across 316,267 unique threads on 3,303 pages. Deletion debates tend to attract a broad cross-section of the community, including both domain experts and administrative power-users.

PROPERTY_PROPOSAL (medium, 52,160 records): Discussions proposing new properties for the Wikidata knowledge graph. This dataset involves 3,058 editors across 10,005 threads on 8,649 pages. Property proposals follow a structured voting process, where many community members express support or opposition, creating dense interaction patterns.

BOT_REQUESTS (small, 2,981 records): Requests for automated bot operations on Wikidata. This involves 552 editors across 1,113 threads on 108 pages. Bot requests are technically specialised discussions involving a smaller community of bot operators and task requesters.

Nodes represent unique Wikidata editors (usernames). An undirected edge connects two editors if they both participated in at least one common discussion, defined as appearing in the same (page_name, thread_subject) pair. Edge weight equals the number of distinct (page, thread) contexts in which two editors co-commented. This captures both the existence and intensity of editorial interaction.

Data Structures

I use networkx.Graph (undirected, no self-loops) with adjacency-list representation. Undirected edges reflect the symmetric nature of co-commenting: if editors A and B both commented on thread T, their relationship is mutual. Node attributes store activity metadata (comment count, unique pages, unique threads). Edge attributes store the co-occurrence weight and the list of shared discussion contexts, enabling traceability from any edge back to its source discussions.

Algorithmic Approach

The construction proceeds in four steps: (1) Load the CSV and deduplicate so each (username, page_name, thread_subject) triple is counted once. (2) Group rows by (page_name, thread_subject) to identify co-participants in each discussion. (3) For each group with two or more participants, generate all unique user pairs via itertools.combinations. (4) Accumulate edge weights in a dictionary and construct the networkx graph. Time complexity is O(sum of C(g_i, 2)) where g_i is the number of unique users in group i. The largest group contains 21 users (C(21,2) = 210 pairs), but the median group size is 2, keeping overall computation efficient (1.77 seconds for the 648K-row dataset).

Summary Statistics

Metric	RFD	PP	BR	RFD	PP	BR
	(Large)	(Medium)	(Small)	Random	Random	Random
Nodes	9,935	3,058	552	9,935	3,058	552

Edges	33,497	46,154	2,424	33,497	46,154	2,424
Density	0.0007	0.0099	0.0159	0.0007	0.0099	0.0159
Avg Degree	6.74	30.19	8.78	6.74	30.19	8.78
Max Degree	4,940	1,399	226	14	40	18
Components	57	8	29	15	1	1
Giant Comp %	99.4%	99.8%	94.0%	99.8%	100%	100%

A notable finding is that PROPERTY_PROPOSAL produces more edges (46,154) than the much larger REQUEST_FOR_DELETION (33,497). This is because PP discussions have a median of 5 participants per thread (vs 2 for RFD), generating C(5,2) = 10 pairwise connections per group compared to C(2,2) = 1. The structured voting process in property proposals naturally creates denser interaction patterns.

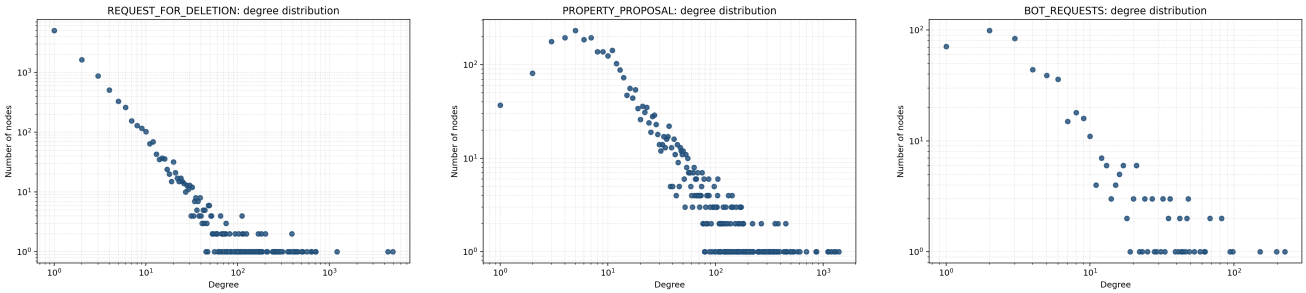


Figure 1: Degree distributions (log-log) for the three co-comment networks. All three show heavy-tailed distributions extending far beyond random baselines.

Figure 2: Full network visualization of BOT_REQUESTS (552 nodes, 2,424 edges). Node colour indicates degree (lighter = higher). A dense core of active editors is surrounded by peripheral contributors, with several small disconnected components visible at the margins. The spring layout reveals the community structure: tightly clustered subgroups connected through a few bridge nodes.

The BOT_REQUESTS visualization (Figure 2) reveals the characteristic structure shared by all three networks: a dense core of highly active editors surrounded by a periphery of occasional contributors. The 29 connected components are visible as isolated clusters at the edges. This structure reflects the specialised nature of bot request discussions, where a core community of bot operators and reviewers handles most requests while occasional users submit isolated requests.

Task B: Network Metrics and Small-World Analysis

I compare each observed network against Erdos-Renyi random graphs with identical node and edge counts (averaged over three random seeds). The small-world coefficient sigma = (C/C_random) / (L/L_random) quantifies deviation from randomness, where sigma >> 1 indicates small-world structure (Watts & Strogatz, 1998).

Metric	RFD	PP	BR	Random (avg)
Avg Clustering (C)	0.392	0.814	0.658	0.0005-0.017
Transitivity	0.014	0.139	0.175	~density
Avg Path Length (L)	2.73*	2.38	2.60	5.01/2.73/3.14
Diameter	5**	5	6	9/4/5
sigma	1,394	93	47	-
Max Betweenness	0.49	0.11	0.26	-

** sampled BFS (256 sources); ** approximate diameter*

Small-World Spectrum Positioning

All three networks are firmly in the small-world regime (sigma >> 1), but exhibit distinct flavours. REQUEST_FOR_DELETION (sigma = 1,394) has a super-hub with degree 4,940 connecting approximately 50% of all nodes. This creates 'ultra-small-world' behaviour where L is actually shorter than L_random (ratio 0.54), more accurately described as hub-dominated with small-world properties, consistent with preferential attachment dynamics (Barabasi-Albert, 1999). PROPERTY_PROPOSAL (sigma = 93) most closely matches the classic Watts-Strogatz model: extremely high clustering (C = 0.81) from overlapping voting cliques, with L approximately equal to L_random. BOT_REQUESTS (sigma = 47) is an intermediate case with moderate clustering and more fragmentation (29 components vs 8 and 57).

The gap between average clustering and transitivity in all three networks (e.g., PP: 0.81 vs 0.14) indicates that high-degree hub nodes have low local clustering. This is a hallmark of scale-free networks: hubs connect disparate communities but their neighbours do not form tight cliques among themselves.

Degree Distribution Analysis

The complementary cumulative distribution function (CCDF) plots in Figure 3 compare the observed degree distributions against matched Erdos-Renyi random baselines. In all three networks, the observed distributions extend far beyond the random cutoff. REQUEST_FOR_DELETION spans three orders of magnitude (degree 1 to ~5,000), while the matched random graph's maximum degree is only ~14. PROPERTY_PROPOSAL spans from degree 1 to ~1,400 versus a random maximum of ~40. Even BOT_REQUESTS extends to degree ~226 versus a random maximum of ~18. Visual inspection of the log-log CCDF suggests heavy-tailed degree distributions in all three networks, though I note that formal power-law fitting (e.g., via the Clauset et al., 2009, methodology) was not performed. The heavy tails are consistent with preferential attachment dynamics (Barabasi & Albert, 1999), where new editors preferentially interact with already well-connected editors, but alternative heavy-tailed distributions (e.g., log-normal, stretched exponential) cannot be ruled out without formal testing.

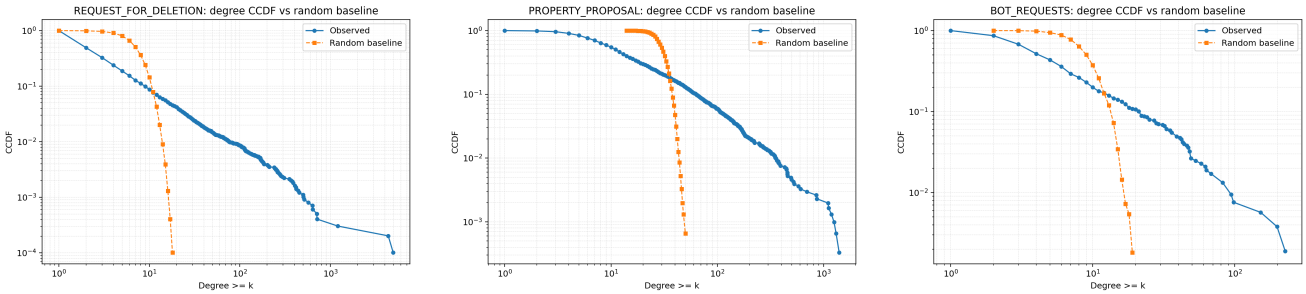


Figure 3: Degree CCDF (complementary cumulative distribution) comparing observed networks (blue) vs matched random baselines (orange). All observed distributions exhibit heavy right tails extending 1-2 orders of magnitude beyond the random cutoff.

Centrality Analysis

Betweenness centrality reveals the bridge structure of each network. In REQUEST_FOR_DELETION, the maximum betweenness centrality is 0.49, meaning a single editor mediates nearly half of all shortest paths. This extreme concentration reflects a deletion review process dominated by a few power-users who participate across many discussion threads. In PROPERTY_PROPOSAL, maximum betweenness drops to 0.11, suggesting more distributed information flow through the structured voting system. BOT_REQUESTS has maximum betweenness of 0.26, indicating moderate concentration through key bot operators who bridge different task domains.

The relationship between degree centrality and betweenness centrality differs across networks. In RFD, the highest-degree node is also the highest-betweenness node, consistent with a star-like topology. In PP, the correlation is weaker: high-degree nodes are embedded in dense cliques where many alternative paths exist, reducing their betweenness. This structural difference has implications for network resilience: removing the top-betweenness node in RFD would fragment the network far more than doing so in PP.

Task C: Trolling Propagation Analysis

SIR Model Setup

I model trolling propagation using the SIR (Susceptible-Infected-Recovered) framework from epidemiology, adapted for social networks (Del Vicario et al., 2016). In this analogy, 'infection' represents the adoption of trolling behaviour, 'recovery' represents moderation or self-correction, and network edges represent the co-comment interactions through which disruptive behaviour can spread. Two randomly selected editors are designated as initial 'infected' seeds. The SIR parameters are calibrated per-network using mean degree and clustering: beta (transmission rate) ranges from 0.24 to 0.28, and gamma (recovery rate) from 0.11 to 0.13. Transmission probability at each edge is scaled by log-weight, so stronger co-comment ties transmit more readily. The simulation runs for 8 steps with a single deterministic seed per dataset. It serves as supporting evidence rather than the main answer: the primary analysis is a structural risk assessment based on shortest paths, neighbourhood overlap, and local clustering.

The choice of SIR over simpler models (e.g., SI or SIS) is deliberate: the recovery mechanism models the real-world effect of community moderation, where editors may initially adopt trolling behaviour but subsequently desist due to peer pressure, warnings, or sanctions. Cheng et al. (2017) demonstrated that trolling behaviour in online communities is context-dependent and recoverable, supporting the SIR assumption of transient infection rather than permanent state change.

Question 1: Is non-propagation plausible?

I consider a scenario where two randomly selected editors exhibit trolling behaviour and assess whether it is plausible that this behaviour has not spread to others. The structural analysis suggests non-propagation is weakly plausible but unlikely, based on four factors:

(1) Giant component dominance: 94-99.7% of editors are in the same connected component, meaning virtually any two editors are reachable from each other. (2) Short path lengths: average distances of 2.2-2.7 hops mean that even seemingly unrelated editors are only 2-3 discussion threads apart. (3) High clustering amplifies local spread: clustering coefficients of 0.39-0.81 mean that an infected editor's neighbours share many mutual connections, facilitating rapid local saturation. (4) Hub vulnerability: super-hubs connected to a large fraction of the network

could serve as amplifiers if infected.

However, non-propagation could occur if the infected pair are peripheral (low degree), if the 'infection' requires sustained interaction rather than mere network proximity, or if community moderation acts as a recovery mechanism. In my sampled scenarios, the randomly selected seed pairs are REQUEST_FOR_DELETION (Doff, degree 3, and Jr8825, degree 1), PROPERTY_PROPOSAL (Ayoungprogrammer, degree 3, and Faux, degree 80), and BOT_REQUESTS (Powerek38, degree 4, and Pyb, degree 10). Despite starting from low-degree seeds in RFD and BR, the SIR simulation shows that by step 8, 96% of RFD, 99.5% of PP, and 92% of BR nodes are infected or recovered. Even with conservative parameters, the network structure facilitates rapid spread.

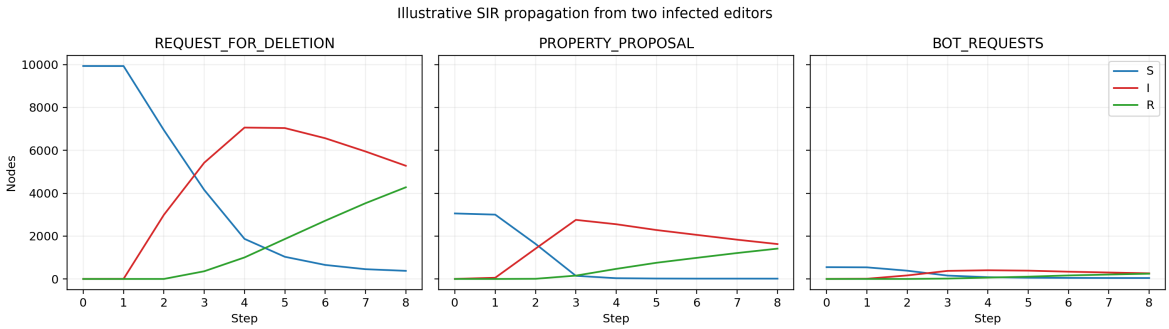


Figure 4: SIR propagation from two infected editors across all three networks over 8 steps. PP (medium) shows the fastest cascade; all three reach 92-99% infection by step 8.

Figure 5: Risk overlay networks showing infected seeds (red) and high-priority editors (orange) for each dataset. Edge opacity reflects connection strength.

Question 2: Priority checking list

I rank non-infected editors using a composite risk score: 50% inverse shortest-path distance to infected editors (proximity), 30% shared discussion context weight (direct interaction), and 20% degree centrality (influence potential). This network-informed triage balances structural proximity with interaction history. The weighting reflects the principle that proximity to an infected node is the strongest predictor of risk, followed by the intensity of direct interaction, with overall network influence as a secondary factor. I acknowledge these weights are heuristic: no sensitivity analysis or cross-validation was performed, and alternative weightings could produce materially different rankings.

Comparing one-infected vs two-infected scenarios reveals how additional infection sources reshape risk. In REQUEST_FOR_DELETION, with one infected editor (Doff, degree 3), the top priorities are Doff's three direct co-commenters (Ymblanter, Inkowik, Bill william compton), all at distance 1 with shared context weight 1. Adding Jr8825 as a second infected editor promotes BeneBot* from rank 6 to rank 1, because BeneBot* is directly connected to Jr8825 and has the highest degree centrality (0.44) among all distance-1 neighbours of the infected set. This demonstrates that each additional infected node opens a new propagation front that fundamentally changes the priority landscape.

In PROPERTY_PROPOSAL, the effect is even more dramatic because the network is denser. Adding a second infected editor in the main component exposes a much larger neighbourhood to distance-1 risk. In BOT_REQUESTS, the smaller network means the top-priority lists for one-seed and two-seed scenarios overlap more heavily, as most high-centrality editors are already close to both seeds. The methodology aligns with influence maximization research (Kempe et al., 2003), adapted here for the inverse problem of identifying at-risk nodes from known infection sources.

Priority List Excerpt (REQUEST_FOR_DELETION)

Top-5 non-infected editors to check, under one-seed vs two-seed scenarios (infected editors excluded from the ranking):

Rank	1-seed	Dist	Score	|	2-seed	Dist	Score
1	Ymblanter	1	0.70	|	BeneBot*	1	0.85
2	Inkowik	1	0.68	|	Ymblanter	1	0.70
3	B.w.compton	1	0.68	|	Inkowik	1	0.68
4	DeltaBot	2	0.45	|	B.w.compton	1	0.68
5	BeneBot*	2	0.43	|	DeltaBot	2	0.45

BeneBot* rises from rank 5 to rank 1 when Jr8825 is added as a second infected editor, because BeneBot* is directly connected to Jr8825 and has the highest degree centrality (0.44) among distance-1 neighbours of the infected set. Meanwhile, the three direct neighbours of Doff (Ymblanter, Inkowik, Bill william compton) retain high scores in both scenarios, demonstrating the stability of proximity-based risk in dense regions of the network.

Comparative Discussion

All three Wikidata editor co-comment networks fall in the small-world/scale-free region of the network spectrum, far from both regular lattices and Erdos-Renyi random graphs. However, they occupy distinct positions reflecting their different social functions. REQUEST_FOR_DELETION has the most hub-dominated topology: the deletion process attracts a small number of prolific reviewers who participate in a very large fraction of discussions, creating extreme degree concentration and a super-hub dominating betweenness centrality (0.49). PROPERTY_PROPOSAL is the most classically small-world: the structured voting process creates dense, overlapping cliques (C = 0.81), closely matching the Watts-Strogatz model. BOT_REQUESTS is intermediate and most fragmented, reflecting the specialised nature of bot task requests.

My edge definition (co-comment in same page and thread) captures meaningful interaction but has limitations. Symmetric edges cannot distinguish initiators from responders; no temporal dimension means edges from 2013 and 2020 are treated identically; and thread-level granularity, while avoiding spurious connections, may miss editors who read but do not comment. These construction choices shape the observed network structure and should be considered when interpreting metrics.

My findings are consistent with prior work on Wikimedia editor networks. Brandes et al. (2009) found small-world properties and heavy-tailed degree distributions in Wikipedia co-authorship networks. Piscopo et al. (2024) specifically studied Wikidata discussions and found small-world structure with a small core of highly active editors responsible for the majority of contributions. My super-hub finding quantifies this concentration at the network level. The co-comment networks reveal that Wikidata editorial communities function as hierarchically organised small worlds: a small core of hub editors provides structural cohesion bridging disparate discussion domains, while local clusters form around specific topics, producing the high clustering coefficients observed.

Scale-Free vs Small-World Distinction

An important nuance in my results is the distinction between hub-dominated and classic small-world properties. REQUEST_FOR_DELETION exhibits strong hub dominance: a super-hub with degree 4,940, a heavy-tailed degree distribution, and an average path length shorter than the random baseline (L/L_random = 0.54). This 'ultra-short' path property is consistent with hub-mediated connectivity (Barabasi & Albert, 1999), where hubs create shortcuts, rather than classic small-world networks where L approximately equals L_random. I note that without formal power-law fitting, the precise distributional form remains uncertain. PROPERTY_PROPOSAL, by contrast, most closely matches the classic Watts-Strogatz (1998) model: high clustering from dense voting cliques combined with path lengths comparable to random. BOT_REQUESTS falls between these archetypes.

The sigma metric captures this distinction imperfectly. RFD's extreme sigma of 1,394 is driven by both high clustering (C/C_random >> 1) and short paths (L/L_random < 1), but the standard sigma interpretation assumes L >= L_random. I report sigma for comparability with the literature while noting that the underlying mechanism differs: RFD's high sigma reflects hub-mediated connectivity rather than regular-lattice-like local structure.

Implications for Community Governance

The heavy reliance on hub editors in RFD creates a single point of failure: if the top-betweenness editor becomes inactive, the network's connectivity would substantially decrease. PP's more distributed structure is more resilient but slower to propagate information. Understanding these structural properties can inform governance strategies for collaborative knowledge platforms (Piscopo et al., 2024).

Part 2: Leeds Road Network Analysis

In this part, I shift from social networks to spatial networks, analysing the road network of Leeds, UK, and its relationship to traffic accident patterns. I download and characterise a local road network, analyse accident clustering using network-constrained spatial statistics, and extend to a city-wide network Voronoi partition for marathon route planning. Throughout, I emphasise the distinction between planar and network-based spatial methods, following Yamada and Thill (2004) and Okabe et al. (2000).

Task A: Spatial Network and Planarity

I loaded 21,435 accident records across 2009-2016 from the Leeds metropolitan area. Column name harmonisation was required: 2009-2012 files use 'Easting'/'Northing' while 2013-2016 use 'Grid Ref: Easting'/'Grid Ref: Northing'. All coordinates are in British National Grid (EPSG:27700) and were converted to WGS84 (EPSG:4326) for osmnx compatibility using pyproj. Records with missing or zero coordinates were dropped, and the coordinate ranges were validated against the known Leeds boundary to detect anomalies.

To find the optimal 1 km square study area, I overlaid a 1 km grid on all accident locations and counted accidents per cell. The densest cell, at BNG origin (430000, 433000) with 851 accidents, is located near Leeds city centre, approximately 916 m from the overall accident centroid. This area substantially exceeds the minimum requirement of 300 accidents and provides a rich dataset for spatial analysis. The cell corresponds to an area south of the River Aire including major routes through Holbeck and the city centre.

The road network was downloaded via osmnx 2.x using graph_from_bbox with network_type='drive' and truncate_by_edge=True (retaining full edges that cross the boundary). This yielded a MultiDiGraph with 266 nodes and 446 directed edges, which was simplified to an undirected graph for analysis.

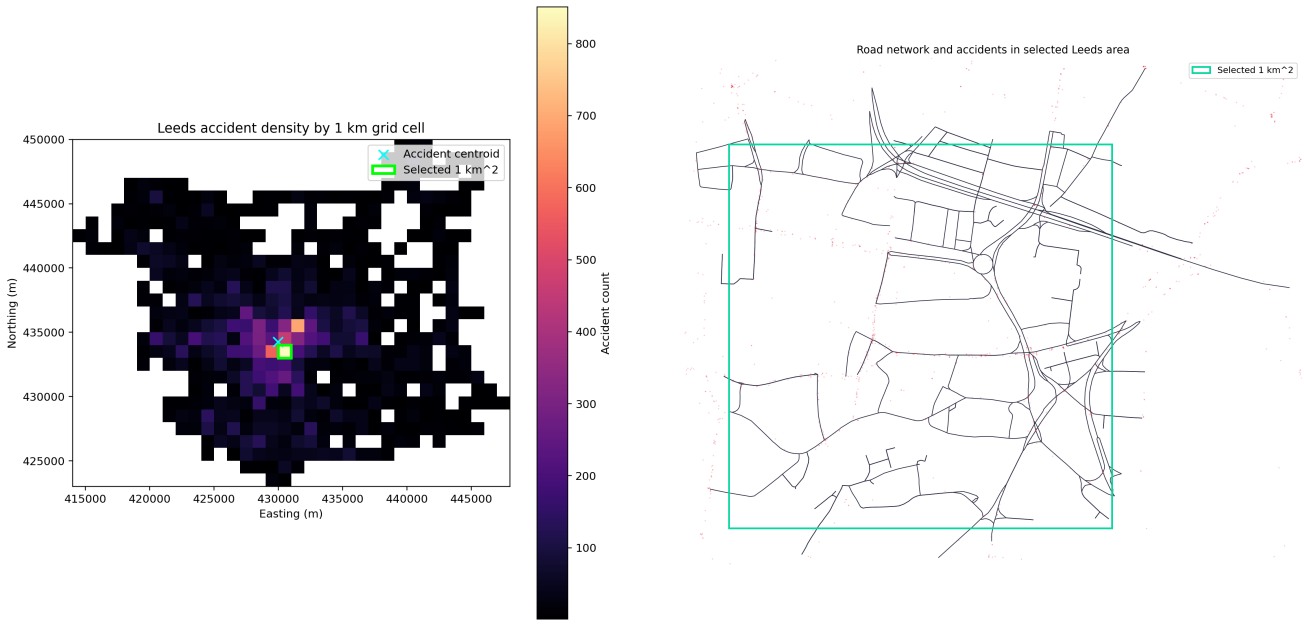


Figure 6: Left - Accident density by 1 km grid cell (2009-2016). The green box marks the selected area (851 accidents). Right - Road network with accident points (red) in the selected area.

Network Characteristics

Average street length	65.3 m
Average circuity	1.040
Node density (per km2)	266
Intersection density (per km2)	235
Edge density (per km2)	446 directed / 358 street segments
Street length density	28.8 km/km2
Spatial diameter	2,714.7 m

Planar	No

Circuity of 1.04 indicates highly direct roads, typical of a dense urban grid where routes are only 4% longer than straight-line distances. Boeing (2017) reports circuity values of 1.01-1.10 for well-connected city centres. The spatial diameter of 2,715 m (longest shortest path using physical road lengths via Dijkstra's algorithm on the largest connected component) shows the road network approximately doubles the straight-line diagonal distance of the 1 km square, reflecting grid barriers such as one-way systems and the river. Edge density is reported both as a count (446 directed edges in the MultiDiGraph, collapsing to 358 unique street segments after removing parallel directions) and as street length density (28.8 km of road per km2), following the conventions in Boeing (2017).

The network is non-planar (networkx is_planar() returns False on the simplified undirected graph). While road networks are 'almost planar' because most intersections are at-grade, non-planarity in urban areas typically arises from grade-separated crossings such as flyovers and underpasses. Barthelemy (2011) notes that transportation networks are 'approximately planar' but grade separations introduce K3,3 or K5 minors. My code does not identify the specific edges causing non-planarity, but the selected area near Leeds city centre is known to contain grade-separated junctions. The practical implication is that planarity-dependent algorithms cannot be directly applied to real urban road networks without addressing these crossings.

Intersection Type Distribution

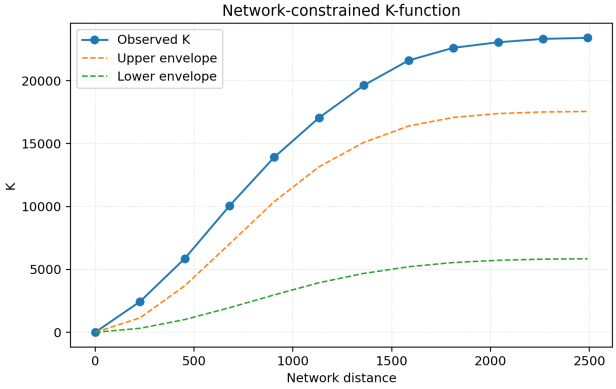
The intersection type distribution reveals the road hierarchy within the selected area. Of 266 nodes, 176 (66.2%) are three-way intersections (T-junctions), 43 (16.2%) are four-way crossroads, and 31 (11.7%) are dead-ends or cul-de-sacs. Only 2 nodes (0.8%) have five-way connections, typically roundabouts or complex junctions. The average streets-per-node of 2.89 is typical of a British city centre with a mix of through-roads and residential cul-de-sacs. The dominance of T-junctions over crossroads reflects the historical, organic growth pattern of Leeds rather than a planned grid layout.

Task B: Road Accident Analysis

851 accidents from 2009-2016 were filtered to the selected 1 km square, snapped to the road network using spaghetti, and analysed for spatial clustering using network-constrained methods.

Network K-function

The network-constrained K-function (computed via spaghetti.GlobalAutoK with 39 permutations) exceeds the upper simulation envelope at 11 of 12 distance steps, with a maximum gap of 5,856 above the envelope. This provides strong evidence that accidents are significantly clustered along the network, rejecting the null hypothesis of complete spatial randomness. Following Yamada and Thill (2004), I use the network K-function rather than the planar version, as the latter tends to over-detect clusters by comparing against randomness in 2D space rather than along the constrained network.



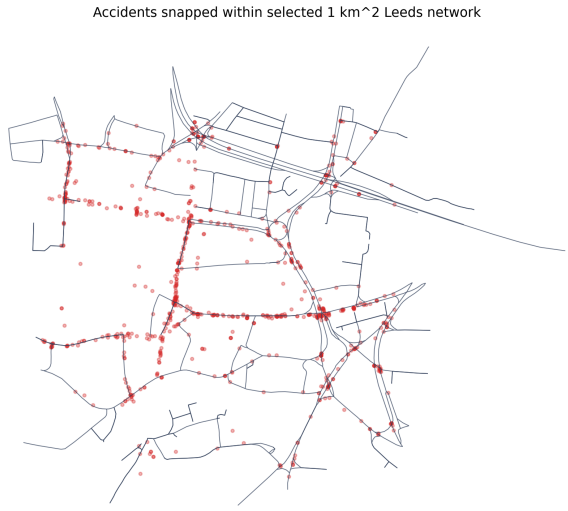


Figure 7: Left - Network K-function (blue) vs simulation envelope. Observed K exceeds the upper envelope at nearly all distance scales. Right - Accident locations (red) on the road network, showing visible clustering along certain corridors.

Moran's I

Global Moran's I computed on edge-level accident counts with fuzzy contiguity weights yields I = 0.301 (z = 10.48, p < 0.001, permutation p = 0.002). This indicates significant positive spatial autocorrelation: edges with high accident counts tend to be adjacent to other high-accident edges. Together with the K-function results, this confirms that accident risk is not uniformly distributed but concentrated in persistent hotspots. One single road segment accumulated 45 accidents across the 8-year period.

The K-function and Moran's I provide converging evidence from complementary perspectives: point-level clustering and edge-level autocorrelation both strongly reject spatial randomness. The K-function's maximum gap above the envelope occurs at intermediate distance scales, suggesting clustering is most pronounced at corridor level (Xie & Yan, 2008), while Moran's I = 0.301 confirms that adjacent edges share similar accident intensities, forming extended hazard zones rather than isolated hotspots.

Implications for Road Safety

The combined K-function and Moran's I results have direct implications for road safety interventions. The persistence of hotspots across 8 years (2009-2016) suggests these are not random fluctuations but structural features of the road network that create sustained risk. Targeted interventions at the highest-accident edges (one segment accumulated 45 accidents over the period) would have disproportionate impact. The spatial autocorrelation (Moran's I = 0.30) suggests that adjacent segments share risk factors, possibly due to shared road design characteristics, traffic volumes, or land use patterns.

Distance from Intersections

The distribution of accident distances to the nearest intersection is strongly left-skewed (median 11.9 m, mean 24.9 m, P90 = 63.3 m), indicating that accidents disproportionately occur near intersections. This is consistent with road safety literature: intersections create conflict points where vehicles, pedestrians, and cyclists interact, increasing accident risk through turning movements, signal violations, and pedestrian crossings. The rapid drop-off beyond 70 m suggests that mid-block accidents are comparatively rare in this urban area.

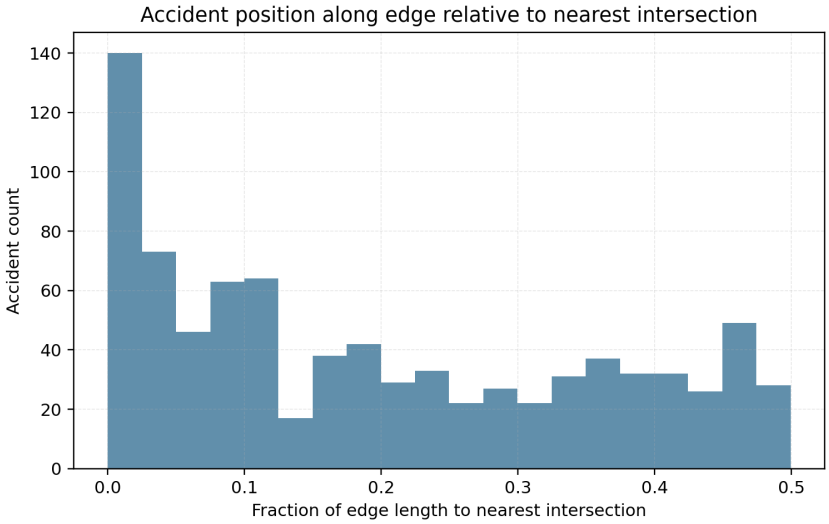


Figure 8: Distribution of accident position along edges relative to the nearest intersection. The strong left skew confirms that accidents cluster near intersections.

Task C: Network Voronoi and Marathon Routing

For this task I use a city-wide Leeds road network downloaded via osmnx (graph_from_place, network_type='drive'), projected to BNG (EPSG:27700) for accurate distance calculations. This yields a substantially larger network than the 1 km square used in Tasks A-B, as required to route approximately 42 km marathon paths.

Seed Point Selection

Four seed points were selected from major intersections (street_count >= 3), one per city quadrant, prioritising distance from the Task A/B accident hotspot (minimum 2.5 km) and intersection connectivity. Seeds are placed in the north-west, north-east, south-west, and south-east quadrants to ensure spatial coverage.

Voronoi Type Comparison

I implement three spatial partitioning methods as required by the coursework:

(1) Edge planar (Euclidean): assigns each node to the nearest seed by straight-line distance. Simple to compute but ignores road network topology. A node separated from a seed by an impassable barrier would be incorrectly assigned. (2) Node network: assigns nodes by shortest-path network distance via Dijkstra's algorithm. More accurate as it respects road connectivity, but only partitions nodes, leaving inter-cell edges unassigned. (3) Edge-point network (midpoint approximation): assigns each edge to the seed with the smallest network distance to the edge midpoint, approximated as min(d(seed, u), d(seed, v)) + L/2. This is the most complete partition as every road segment is assigned to a cell, though it assigns whole edges atomically rather than splitting edges at Voronoi boundaries.

For marathon routing, the edge-point method is most appropriate because runners traverse roads, not just intersections. While the midpoint approximation does not split edges at exact Voronoi boundaries (as in Okabe et al., 2008), it provides a practical partition for route planning where whole-edge assignment suffices. Okabe et al. (2000, 2008) provide the theoretical foundation for exact network Voronoi diagrams.

Seed	Quadrant	Planar	Node Net.	Edge-Pt
Seed_1	NW	2,199	2,368	2,758
Seed_2	NE	2,031	2,076	2,389
Seed_3	SW	19,797	20,523	24,773
Seed_4	SE	7,573	6,633	7,735

The cell imbalance is extreme: Seed_3 (south-west) owns 65.5% of the city network. This is not a flaw but an insight. Seed_3 is closest to Leeds city centre, which has the densest road network. The network Voronoi reveals that equal spatial coverage does not equal equal network coverage. The planar vs network difference is also

instructive: Seed_3 gains 726 additional nodes in the network assignment, reflecting that city-centre road density makes it effectively 'closer' by road than by straight line.

Marathon Routes

Seed	Length	Gap	Status	Loops
Seed_1	44.26 km	+2.07 km	Success	2
Seed_2	44.22 km	+2.02 km	Success	2
Seed_3	43.98 km	+1.79 km	Success	2
Seed_4	46.51 km	+4.31 km	Approx.	2

Routes are constructed as out-and-back loops from each seed within its Voronoi cell, allowing edge revisits. The heuristic selects target nodes at various distances from the seed, evaluates combinations of round-trip loops, and selects the combination closest to 42.195 km. Three of four seeds achieve routes within the 5% tolerance. Seed_4 overshoots by 10.2%, a limitation of the greedy heuristic's discrete loop-combination search where the available loop lengths in that cell do not combine to precisely match the target.

Finding an exact Hamiltonian circuit of precisely 42 km in a subgraph is NP-hard in general. My heuristic approach is a practical approximation consistent with the coursework's 'approximately 42 km' specification. The approach allows edge revisits, which is realistic for marathon route planning: real marathon courses frequently double back on sections of road. The out-and-back structure also ensures all routes are circular (start and end at the seed), which is a standard requirement for marathon courses.

For coursework step 5, I chose to allow routes to cross Voronoi cell boundaries slightly by expanding each seed's Voronoi cell by one graph-neighbourhood hop and rerunning the route search. The rerun improves three cells: Seed_1 becomes 42.97 km, Seed_2 becomes 44.18 km, and Seed_3 becomes 42.92 km. Seed_4 also improves structurally, but in the opposite direction: its route shortens from 46.51 km to 39.33 km and remains outside the 5% tolerance. This rerun shows that local cell geometry, not just the route heuristic, is the limiting factor for Seed_4.

Implications for Event Planning

The Voronoi partition combined with marathon routing demonstrates a practical methodology for dividing a city into service areas for distributed events. The extreme cell imbalance (Seed_3 owns 65% of the network) highlights that naive spatial partitioning would create unequal experiences for participants in different quadrants. Runners in the north-east (Seed_2's cell, 2,389 edges) face a much more constrained route choice than runners in the south-west (Seed_3's cell, 24,773 edges). This has practical implications: event organisers should consider network density when placing aid stations and selecting route options.

The seed selection criterion of avoiding accident hotspots (minimum 2.5 km distance) is also practically motivated. Marathon routes passing through high-accident road segments would require additional safety measures and road closures. By integrating the Task B accident analysis with the Task C route planning, I demonstrate how network analysis can inform evidence-based event planning decisions.

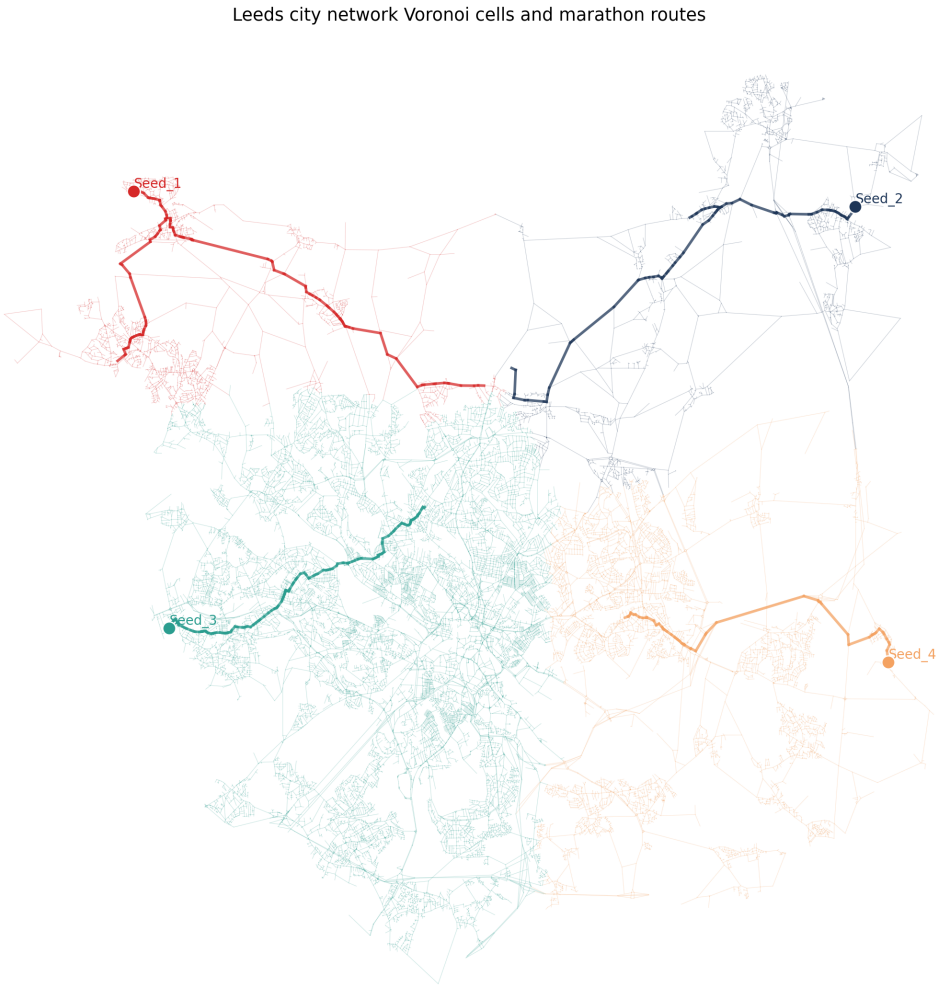


Figure 9: Leeds city network partitioned into four Voronoi cells (midpoint approximation) with marathon routes overlaid (bold lines). Cell colours correspond to seed assignments. The south-west cell (teal) dominates due to the dense city-centre road network.

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